Assignment 3

Name: Afifa Kaleem Butt
Section: 8D
Subject: Artificial Neural Networks
Sap ID: 70128235

Submitted to: Ma'am Dua Nadeem

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The University of Lahore

Q) CNN Architecture:

Suppose the input is a 32 x 32 x 3 RGB image

Layer 1 : Convolutional Layer

- Filters : 16 filters of size 3x3
- Stride : 1
- Padding : 1
- Activation function : ReLU

Ans:- ....

★ Input Layer

Input : 32x32 image with 3RGB channels

Parameters : None. There is no learning at this stage

★ Convolutional Layer:

- Input : 32x32x3
- Filter : 3x3x3 , 16 filters , Padding : 1 , Stride:
- Output Size :

$$= \frac{32 - 3 + 2(1)}{1} + 1$$

$$= 32 - 3 + 2 + 1$$

$$= 32$$

$$= 32 \rightarrow 32 \times 32 \times 16$$

Parameters :

$$(3 \times 3 \times 3 + 1) \times 16 = 448$$

Activation : ReLU

★ Pool 1: Max Pooling Layer

Filter : 2x2, Stride : 2,

Output : $\frac{32-2}{2}+1 = 16$

$= 16 \rightarrow 16 \times 16 \times 16$

Parameters : 0

Activation : None. (Pooling Layer)

★ Conv 2 : Convolutional Layer

- Input : $16 \times 16 \times 16$
- Filter : $3 \times 3 \times 16$, 32 filters, Padding : 1, Stride : 1
- Output : $\frac{16-3+2(1)}{1}+1$

  $= 16 \rightarrow 16 \times 16 \times 32$
- Parameters :

  $(3 \times 3 \times 16 + 1) \times 32$

  $= 4,640$

Activation function : ReLU

★ Pool 2 : Max Pooling layer

Filter : 2×2, Stride 2

Output : $\frac{16-2}{2}+1$

$= 8$

$= 8 \rightarrow 8 \times 8 \times 32$

Parameters : 0

Activatation : None.

★ Conv 3: Convolutional Layer

Input 8 × 8 × 32

Filter : 3 × 3 × 32, 64 filters, Padding : 1, Stride : 1

- Output : $\frac{8 - 3 + 2(1)}{1} + 1 = 8 \rightarrow 8 \times 8 \times 64$

Parameter : $(3 \times 3 \times 32 + 1) \times 64 = 18{,}496$

Activation : ReLU

★ Conv 4 : Convolutional Layer

Input : 8 × 8 × 64

Filter : 3 × 3 × 64, 128 filters, Padding : 1, Stride = 2

Output : $\frac{8 - 3 + 2(1)}{2} + 1 = 4 \rightarrow 4 \times 4 \times 128$

Parameters : $(3 \times 3 \times 64 + 1) \times 128 = 73856$

Activation : ReLU

★ FC 1 : Fully Convolutional Layer

Input : Flattened 4 × 4 × 128 = 2048

Output units = 256

Parameters : (2048 ~~× 1~~ 256) ~~×~~ 256 = ~~524,544~~ ~~5898241~~

524, 544

Activation : ReLU.

★ FC 2 : Output Layer

- Input : 256 units
- Output : 10 units (10 - class classification)
- Parameters : (256 × 10) + 10 = ~~2570~~ ~~25800~~ 2570
- Activation : Softmax.

Formula & Steps Explained

1. Feature Map Size (for Convolution and Pooling Layers)

we use this standard formula for the output size of a convolution or pooling layer:

Output Size: $\frac{W - F + 2P}{S} + 1$

where: W = input width or height
F = filter size (kernel size)
P = padding.
S = stride

This formula is applied seprately for width and height (they are usally equal for small square images), and the depth (channels) is determined by the number of filter's in that layer.

2. Number of Parameters (for Convolution Layers)

The number of learnable parameters in a convolutional layer is given by:

Parameters = $(F \times F \times \text{Input Channels} + 1) \times$ Number of Filters.

Explanation: $F \times F \times$ Input Channels: weights per filter. + 1: adds a bias term per filter. x Number of Filters: total filter's in the layer.

3. Feature Map Size (for Pooling Layers)

Pooling layers donot have parameters. They just downsample the input. The output size for pooling is calculated with the same formula.

Output Size $= \frac{W-F}{S} + 1$ (Assume padding $P=0$ by default in most padding/pooling layers).

4. Parameters in Fully Connected Layers:

Fully connected (dense) layers have parameters calculated a

Parameters = (Number of Input Units + 1) x No. of Output Units

- +1 accounts for the bias term in each output unit
- Input is flattened if coming from convolutional layers.

Example Applications of Formulas:

Conv 1 :

- Input : 32x32x3, Filters : 16, Filter Size : 3x3, Padding : 1, Stride : 1.
- Output size : $\frac{32-3+2(1)}{1} + 1 = 32 \Rightarrow 32\times32\times16$
- Parameters : $(3\times3\times3+1) \times 16 = 448$

FC 1 :

- Input units : $4\times4\times128 = 2{,}048$, Output units : 256
- Parameters : $(2{,}048+1)\times256 = 524{,}544$.

Layers	Feature Map Size	# Parameters	Activation
Input	32x32x3	0	–
Conv1	32x32x16	448	ReLU
Pool1	16x16x16	0	–
Conv2	16x16x32	4640	ReLU
Pool2	8x8x32	0	–
Conv3	8x8x64	18,496	ReLU
Conv4	4x4x128	73,856	ReLU
FC1	1x256	524,544	ReLU
FC2	1x10	2,570	Softmax